

Self-Modeling to Promote Social Initiations With Young Children With Developmental Disabilities

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Abstract

Video self-modeling (VSM) uses a method called *feedforward* to provide children the opportunity to view themselves as they perform in a more advanced or appropriate manner than they do presently. Typically, this is accomplished through the careful editing of videos. Studies on VSM and social skills with children on the autism spectrum ages 3 to 4 years have produced mixed results. In this study, a single-subject multiple-baseline design across four children (three on the autism spectrum and one with Down syndrome) with a mean age of 4 years 2 months was used to determine whether VSM would facilitate social initiations. This study was meant to be a replication of a study published in 2012. Positive changes were seen for all four children. The relationship between age and VSM efficacy is discussed along with other factors that may influence VSM outcomes with young children.

Keywords

Autism, self-modeling, video

There has been much attention given to video-modeling interventions for children with autism over the past decade (Buggey & Ogle, 2012). A substantial proportion of the video-modeling research with children with autism involves video self-modeling (VSM; for example, Bellini, Akullian, & Hopf, 2007; Buggey, 2012; Wert & Neisworth, 2003) in which children view edited videos of themselves performing appropriately or at an advanced level. Dowrick (1983) coined the term *feedforward* for the process of obtaining information by viewing positive future functioning of the observer. Interestingly, the majority of studies investigating the efficacy of self-modeling has been carried out with children with autism (e.g., Bellini et al., 2007; Buggey, 2005, 2012; Delano, 2007).

There are several theoretical models that might factor into the established efficacy of VSM and especially VSM use by children with autism. VSM takes advantage of the obvious similarities of the model and the viewer that Bandura (2001) found to be a key component of model efficacy. Bandura's research revealed that the best models were those most similar to the observer in all characteristics including appearance, age, and ability. Likewise, Bandura's studies on the importance of self-efficacy, the belief that one can be successful at a task, found a high correlation between self-efficacy and success. VSM may provide a significant boost to one's self-efficacy by providing an individual with visual evidence (memory) of successful performance, and there is

no way to get a model more similar to the viewer. Other factors relate to the visual nature of the medium. Children with autism tend to be visual learners (e.g., Bauman, 1999; Garretson, Fein, & Waterhouse, 1990; Schiavo, Tressoldi, & Martinez, 2005). This, combined with the fact that there are no social obligations when viewing a model on a screen as opposed to in vivo peers and adults, may add to the strength of the visual learning aspect of VSM.

The number of studies using video modeling in all forms with persons with autism is relatively small, although growing. The studies tend to deal predominantly with adult and peer modeling, with a sizable minority addressing VSM. Bellini and Akullian (2007) conducted a meta-analysis of studies using video modeling with children with autism spectrum disorder (ASD). They found 23 studies including eight that evaluated VSM use with children with autism. Findings across studies revealed that acquisition, generalization, and maintenance of skills were strong for all forms of video

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modeling. VSM results were equivalent or better than other forms of modeling. Bellini and Akullian concluded that all forms of video modeling, including self-modeling, met the Council for Exceptional Children's criteria for research-based practices (Horner et al., 2005).

The number of studies where preschool children with autism served as participants is limited. In fact, only six were identified in the search of the literature for the present study with a total of 13 participants (i.e., Bellini et al., 2007; Buggey, 2005, 2012; Buggey, Hoomes, Sherberger, & Williams, 2012; Litras, Moore, & Anderson, 2010). Wert and Neisworth (2003) used VSM to train spontaneous requesting in four preschoolers with autism. The children were trained to request items via a discrete trial method; however, the resulting requests were rote, and there was no generalization to spontaneous requesting without prompting. A video was created with the prompts removed and only the best examples included. Results, collected using a multiple-baseline design across individuals, indicated that all four participants showed substantial gains. The gains in mean production of spontaneous requests ranged from 800% to 1200%.

Buggey (2005) worked with a 4-year-old boy on the autism spectrum to try to reduce aggressive (pushing) behavior and to increase responses to questions and verbal initiations. A multiple-baseline design across three behaviors was used for evaluation purposes. A decrease in pushing to almost zero occurred and gains were reported for both verbal behaviors. The video addressing pushing depicted replacement behaviors such as touching appropriately and compliance with teacher requests.

Bellini et al. (2007) examined the effects that VSM had on the social engagement of two 4-year-olds with ASD. The authors stressed the importance of obtaining data in natural environments and accordingly recorded percentage of time engaged with peers during normal play times in the classroom. The videos were constructed using footage from brief interactive periods captured in the classroom. They found that following the children's viewing of their videos the percentage of time spent engaged in play with peers increased from 3% to 43% and 6% to 24%, respectively. As in other studies, a strong maintenance effect was seen following withdrawal of VSM.

Buggey et al. (2012) worked with four children, three of whom just turned 4 and one who was 3 years 10 months at the outset of the study. All were diagnosed with moderate to severe autism. The dependent variable was initiation with peers on the playground. A multiple-baseline design across children was used to evaluate results. The 2- to 3-min videos were edited so it appeared that the children were interacting frequently and appropriately with their peers. The children watched videos in their classrooms upon arrival at school for eight successive mornings. The results indicated increased initiations and verbalizations as well as positive changes in

several behaviors not specifically targeted in the videos for three of the four participants. The one child exhibiting no changes in behavior was the youngest participant.

Subsequently, Buggey (2012), using similar methods to the previous study, looked at using VSM to promote social initiations with 3-year-olds with autism. None of the children showed any improvement in initiations. Buggey and Ogle (2013) proceeded to investigate whether the same method could be used with typically developing children close to 3 years of age to encourage them to initiate with peers with autism and again found no change in behaviors.

When looking at the ages of the preschoolers in these studies, there appears to be something occurring at the transition from 3 to 4 years that limits effectiveness of VSM, or at least VSM used to facilitate social skills. It is logical to assume that VSM has limits related to the age of the child and intensity and comorbidity features of the autism. Other factors, such as temperament of the child, complexity of behaviors being addressed, ability to attend, or some combination of these factors, may also contribute.

The purpose of the present study was to investigate whether VSM could be used to increase the number of social initiations of children with ASD, and one child with Down syndrome (DS), during playground activities. It was designed to be a replication of the work of Buggey et al. (2012) and Buggey (2012) while focusing on children near the 3- to 4-year-old transition. The children in the present study averaged 4 years 2 months in age. It is hoped that this study will add to our understanding of the use of VSM with preschoolers with autism and other developmental disabilities in the important area of social skill training and will serve to provide information related to possible limitations of the method.

Method

Setting

Three students who were diagnosed with ASD and one diagnosed with DS participated in this study. They attended a private inclusive preschool in a small Southeastern city. The preschool was part of a larger institute that featured a developmental pediatric clinic and a family and child research center. It also served as an observation site for several local colleges. The preschool had an equal number of children with and without disabilities and this ratio was reflected in each classroom. The classrooms were arranged by age, ranging from 6 weeks to 1 year in the youngest classroom to children 4- to 5-plus years in the oldest. Three of the four participants were in one classroom for children who were approximately 3 to 4 years of age, with one child with ASD in a separate room. The classrooms each had 12 children, six of whom were typically developing and six who had disabilities. The children attended 3 to 5 days per week so that actual

population within the rooms varied somewhat. Each class was staffed by a lead teacher, and two teaching partners (aides). A volunteer grandmother was also assigned to each classroom. The teachers had degrees in early childhood education. The assistants' education ranged from high school graduation to college degrees. Children received a full range of support services including speech, physical, and occupational therapies that were delivered within the classrooms when possible. The children in the present study each participated in occupational and speech/language therapies. The classrooms were typical with low mobile storage shelving serving as dividers for centers. Centers varied in content and revolved around a theme that was initiated across classrooms. A Reggio Emilia philosophy was used at the site so meals were in-room and lighting was subdued. The videos were shown in the teachers' offices that were small enclosed areas off the main classroom.

The playground area where filming and data collection were carried out was approximately 10,000 sq. ft with modern, multicolored equipment. Three large structures with elements for climbing, sliding, and physical manipulation dominated the playground with adjoining swing and sand areas. The three main structures were designed for children of different motor development levels and were wheelchair accessible. A brick wall separated the playground from traffic areas and a wooden fence enclosed the rest of the area.

Participants

Much of the diagnostic information on the children was obtained close to their third birthday as they prepared to transition from early intervention to public school preschool services. Certified school psychologists from the local public school system carried out assessments. Parents and teachers assessed the children on several measures including the *Childhood Autism Rating Scale* (CARS; except for the child with DS).

Child 1. Taylor was 4 years 2 months of age at the onset of the study. On the *Bayley Scales of Infant and Toddler Development*, Taylor had a percentile rank of 3 on the Cognitive Scale. Percentile scores on the Social Skills Subscale of the *Preschool and Kindergarten Behavior Scales-2* were similarly depressed with scores of 1.00 in Social Cooperation, Social Interaction, and Social Independence. The *Gilliam Autism Rating Scale* (GARS) was completed by both mother and teacher with an average score of 28 indicative of average probability of autism. The same persons completed the *Adaptive Behavior Assessment System* (2nd ed.) and obtained similar scores in the domains of General Adaptive, Conceptual, Social, and Practical. Percentile scores across subscales and evaluators ranged from 1.00 to 8.00, indicating significant delays in all areas. The assessments were carried out when Taylor was 2 years 10 months,

and the conclusion was that he exhibited the behaviors consistent with a diagnosis of autism.

Observations carried out on the playground by the author revealed a child engaged in solitary play. He most often positioned himself between a bush and the swings, often picking leaves and stripping them down to the veins. When other children approached him, he would raise his head and often talked, although his words were not conversational and almost always referred to movie dialogue. He sometimes produced appropriate responses to adults by citing lines from movies. He did not show interest in the play of others.

Child 2. Ellen was 3 years 9 months of age at the beginning of the study and 2 years 10 months at the time of her last assessment. The *Adaptive Behavior Assessment System* (2nd ed.) was completed by the mother and teacher. She obtained percentile rank scores below 9.00 in the areas of Conceptual, Social, and Practical skills with comparable scores from the teacher and mother. All ratings were indicative of significant developmental delay.

On the playground, Ellen was observed moving into close proximity of peers; however, she never made direct contact or engaged with others. She had developed a reputation for biting and scratching, so typically children would move away from her resulting in anger, which sometimes led to the biting and scratching. Ellen often appeared tired and would sit at a table and put her head down. Although she approached children physically, there were no other apparent initiations verbally or through physical touching that were not seen as inappropriate.

Child 3. Corey was 4 years 3 months old at the outset of the study. A developmental assessment was conducted just prior to his third birthday. The *Bayley Scales of Infant and Toddler Development-III* was administered and a Cognitive Composite Score of 55 (0.1 percentile) was obtained. The *Adaptive Behavior Assessment System* (2nd ed.) was completed by the mother and teacher. Corey obtained percentile rank scores that did not exceed 1.00 by either rater in the areas of Conceptual, Social, and Practical. All ratings were indicative of significant developmental delay.

Standard scores on the second edition of the GARS (GARS-2) indicated stereotypical behavior and delays in Communication and Social Interaction. An Autism index of 89 was obtained, indicating that autism was "very likely." Corey's score on the CARS, completed by the teacher, was 40, a score that is indicative of severe autism. On the playground, Corey was often seen trailing groups of children as they moved about, but he did not initiate verbally or physically (e.g., patting someone on the shoulder, blocking a path). When a child would run by him or there was a sudden movement, he would start, raising his hands and hopping backward. He actually spent the majority of the time participating in this kind of behavior. He did play on

playground equipment especially the swings, slides, and riding the little cars. Corey was verbal and could participate in simple conversations with his teacher.

Child 4. Jonathan was 4 years 0 months old at the beginning of the study. He was nonverbal, but could communicate with four signs. Scores on the GARS and CARS indicated “very likely” and severe autism, respectively. Jonathan had a positive behavior plan due to aggressive behavior. He would sometimes push children, and his peers would often move out of his way if he approached. He actively participated in solo play on the playground. He was very active and would move among the equipment spending as much time as possible on the swings. Jonathan showed interest in toddlers and infants during the rare times that group was simultaneously on the playground. He exhibited very gentle behaviors with this age group, and the teachers of the younger children did not discourage his interest.

Self-recognition is sometimes listed as a prerequisite for VSM success. To determine whether the children could self-recognize, the same method was used that Buggey and his colleagues (2012) noted. Children were allowed to see themselves by using a large view screen on a camcorder rotated so that it was facing the children. Their reactions were noted. All of the children attended well to their images. They stopped what they were doing and maintained eye contact with the screen for over 30 s. Taylor, Corey, and Ellen engaged in making faces or movement while maintaining eye contact. This display of cause–effect behavior aimed at the camera was considered a good indicator of self-recognition. Jonathan smiled at the screen, but did not make noticeable movements to affect the camera image. He maintained eye contact with the screen for more than 30 s; however, it was less clear if he self-recognized.

Prior to the outset of the study, approval was obtained from the university’s internal review board, and the research committee at the institute where the study was carried out. Informed consent was obtained from the parents of the participants, and amendments were added to the children’s *Individualized Education Programs* (IEPs) as a further measure to inform parents.

Procedure

A single-subject, multiple-baseline design across four children consisting of baseline, intervention, and maintenance phases was utilized for this study. A unique element of the design for this research was that it was carried out for almost the entire school year, beginning in early October 2012 and ending in mid May 2013. The first author had an office at the institute, and it was convenient to devote 0.5 hr per day to data collection. Although the intervention phase lasted only 5 days per student, the yearlong observations allowed for unprecedented analysis of baseline and maintenance

phases. Two methods were used to analyze data. The first was visual inspection of graphs and the second was differences in means across study phases.

Dependent variables. For the purpose of this study, the number of physical and vocal social initiations with peers during playground time was designated as the dependent variable. Definitions for elements of the dependent variable were identical to those used in the study by Buggey (2012) and Buggey et al., (2012). *Social initiation* was divided into subcategories of *physical* and *vocal* initiations. Physical was defined as moving into proximity of a peer or group of peers resulting in *engagement*. Engagement was then defined as attending to the peer or the activity of the peer(s) for more than 5 s while within arm’s reach or actually making physical contact with peers. Vocal initiation was defined as either verbal (use of words) or nonverbal (vocalization including laughing, crying, signs, and utterances to gain attention) behavior that was clearly directed at a peer. Because most vocal initiations also required physical proximity, it was decided that vocal initiation would override physical initiation in the collection procedure to avoid duplication of data (i.e., one initiation counting as both physical and vocal). Thus, once a vocalization occurred, no physical initiation was recorded for that specific event. *Initiation* was further defined so that repetitive activities would not carry undue weight. Once a physical initiation was carried out, no additional physical initiation was recorded unless the children disengaged for a period of 10 s or moved to another activity and then returned. For example, if a child engaged in playing catch and repeatedly threw and caught a ball, only the first action was counted. Repetitive vocal events were similarly not scored except for the first event; however, verbalizations (actual words or phrases) that were separated by at least 10 s and not identical to the previous verbalization were noted and counted as separate data points.

Observation and data collection. Two observers were used in this study: the primary author, who is a university professor, and a student, who was in an honors program and in the psychology program at a local university. Another student from the honors program replaced the first observer for the second semester of the study. Time was spent with each observer prior to live data collection to get agreement of occurrences and type of occurrence to a 90% level. Observers had data collection forms on clipboards and moved as close to the participant as possible so that verbalizations could be heard. Because the playground was a site for student observations and projects, the clipboard was a common site on the playground. The observers would interact with children when a child initiated contact. The observation form is available upon request; however, it had a simple design with columns for tallies of behaviors, type of initiation (physical or communicative), time of initiation, and comments. The comments column was used for recording information about the

behavior including a very brief description, relevant antecedents, site, persons involved, and other environmental factors. The author spent considerable time on the playground over the previous 2 years carrying out other observations so most of the children were familiar with him. The other observers spent approximately 2 weeks on the playground prior to collecting baseline data. It took approximately the same time to reach criterion for each of the observers, approximately 1 week.

Observations were carried out on 70 days. Gaps occurred in data collection due to inclement weather, special events at the children's center, and absences of children.

An event-recording approach (Kennedy, 2005) was used for the collection of data. The observers conducted daily, 15-min observations of participants during playground time. Each child was observed at least once per day. A chart was kept of the number of observations per child and the observers only observed the same child simultaneously when inter-observer agreement was being carried out. Thirty percent of the observations carried out in the baseline, 100% during intervention, and 25% during maintenance involved inter-observer agreement. The order of viewing the children was dependent on when the classes came out for their playground time. Playground time was set for 9:30 each morning, but there was considerable variance.

A 90% agreement rate across observers was set as the criterion for acceptable inter-observer agreement. Two weeks prior to beginning the baseline phase of the study, the inter-observer agreement process began. Because the children with autism rarely interacted, the preliminary observation sessions involved typically developing peers or other children with developmental disabilities who were more active with others. The observers then met to review their findings and discuss differences of opinion. The observers continued to collect and compare data until the 90% agreement was achieved over five successive observations. In order for there to be agreement, both observers had to have a similar description of the behavior, the name or description of the child interacted with, the exact site, and a recorded time that was within 2 min of the other observation. This last criterion was set following the period of initial observations. It became obvious that the recording of time was often done slightly into each observation. The observers were focused on the initiation at first before beginning to write. Thus, the exact time of recorded initiation varied slightly. Also, watches were synchronized to the minute, but not the second. The observers did not stand together, rather, they positioned themselves opposite each other with the participant between them. This limited influencing each other by cueing the recording of an event. It did have a drawback, however, as children sometimes disappeared from view of one of the observers. When one observer recorded an event that was likely out of view of the other (the observer noted the behavior might be out of view

or an observer had no recollection of the event), the observation was disqualified and removed from analysis. This only applied when children were on the move, and one of the observers had noted a question as to whether the child was in view of the other observer.

A frequency-ratio approach was used for calculation of total agreement. This involved taking the sum of occurrences recorded by each observer and dividing the smaller total by the larger total and multiplying by 100 (Kennedy, 2005). Identical data collection procedures were used in the baseline, intervention, maintenance, and follow-up phases of the study.

Independent variable. The independent variable was designed to be as close as possible to the one used by Buggey et al., in their 2012 study. A short 2.5- to 3.5-min video was created depicting participants socially interacting with peers, with an emphasis placed on the initiations. Children in the present study rarely initiated socially in an appropriate manner and had limitations in following directions. Thus, video clips had to be compiled that gave the impression of social initiations. Video footage was obtained by prompting peers to interact with the participants on the playground. The children with autism were taken to the playground with one or two peers in the afternoon when no other children were present. The peers were prompted to get the child with ASD to interact with them (e.g., "Take Ellen over to the slide." "Give Jonathan the shovel."). The software involved in the editing process (iMovie™) allowed the researchers to highlight and emphasize appropriate behaviors. For example, a peer would be prompted to hold the hand of the participant as they moved across the playground. The child with autism may only have sustained the hand-holding for a few seconds; however, those few seconds could be shown at different times in the movie or looped and shown multiple times in succession. We were also able to remove initiations carried out by peers making it appear the participant was the one who initiated.

The peers selected to participate were chosen for their maturity. They enjoyed the extra time on the playground and were very cooperative. The mean amount of raw footage taken of each child was 10 min. Consent for participation was obtained from the parents of the peers, and the study was explained to them.

The primary investigator created the videos for the three participants using iMovie® software on an Apple MacBook Pro™ laptop. Clips were edited and arranged to make it appear as if the child with autism was initiating interactions and playing with peers in an appropriate and fun manner. A very flattering still frame image of the child that depicted the target behavior was selected and placed at the beginning of the movie. A voice-over was done on the still frame announcing the child as the star and labeling the behavior (e.g., "This is Ellen's Movie, starring Ellen! Let's watch Ellen playing nicely with her friends."). Even though the children could not read, titles were placed on the still frame to give it a TV look.

Similarly, another flattering still frame was placed at the end of the movie, and a voice-over added, "Good job playing with your friends, Ellen!" An audio clip of children cheering and clapping followed this. The use of an ambient music background was used to eliminate background noise and adult talk including any prompts. When the child and peers were talking or laughing, the volume of the music was turned down and the actual sound track was turned up. iMovie was used for editing because of its simplicity and ready availability to many therapists, teachers, and parents. Video editing, which traditionally has been a complicated process, has been made accessible to anyone with a computer.

The videos were similar in format with some original content in each. Because each of the children liked to swing, they were shown on the three-person tire swing "having fun" with peers. Footage was chosen that best depicted interaction and any indication of less desirable behavior was eliminated. In Ellen's video, a game of catch was depicted. When Ellen got the ball, she tended to hold it close and run away. However, she did roll it when prompted to do so. Eliminating the footage of her running off made it appear that she was engaged in a game of catch. A Frisbee was also used in this scene. Taylor often talked to himself while standing apart from others. Some of his utterances were appropriate for initiations (e.g., "Hello." "Who are you?" "What is that?"). Therefore, we extracted this audio and reinserted it in scenes where Taylor was in proximity of peers, but with his back turned or when he was at a quarter profile, making it appear he was initiating conversation.

All of the videos were 2 to 3 min in length. Dowrick and Raeburn (1995) stated that the optimum length of VSM videos is 2.5 min with times greater than this not producing any differences in effect. Teachers were provided with a DVD of the movie and were asked to show the video to the children upon their arrival at school each morning for 5 days. The researchers had teachers report on the children's reaction to the movie. They stated all four were able to attend to the screen and exhibited a reaction ranging from interest to glee at seeing themselves. At first, Jonathan would not remain for the entire movie but did for the last three showings.

The videos were shot during the baseline phase of each child. Once the first video was created, and it was clear that the filming process had not changed baseline rates of behavior, Taylor began to view his video. The viewing occurred approximately 1 hr prior to recess, which was approximately arrival time at the preschool. Teachers were given strict instruction to not comment on the video although they could praise the student for watching. Taylor continued to view the video for five sessions, and then it was withdrawn. Five viewings were a rather arbitrary limit but was meant to match the number of views in the studies by Buggey and his colleagues (2012). Ellen began viewing her video shortly after Taylor's was withdrawn. This pattern continued for all four students. There was an inordinately long gap between Corey and Jonathan's VSM movie introduction. This was caused by technical difficulties with the video camera and absences.

Materials/equipment. Video was taken with a Sony™ Handycam DCR-TRV22 mini DV camcorder and edited on an Apple™ MacBook Pro™ laptop computer using the built-in iMovie software program. The newer iMovie program allows for cropping film clips, which allowed the team to remove adults from the movie when they came into camera range and to zoom in on the children without losing too much quality. Other features of iMovie that were used in the production of the self-modeling video were titles, transitions between clips, and audio insertions including clapping, music, and voice-overs.

The filming sessions on the playground utilized the material and equipment that were typically present. Along with the three larger pieces of equipment mentioned in the setting section, balls of various sizes, digging toys for sand play, and tricycles and small enclosed plastic cars were used in the videos and were available for use during recess.

Social validity. Information was gained from adults concerning the VSM procedure, its application in the classrooms, its effects on the students, and the feasibility of using VSM in the future. Teachers were also asked about any behavior changes seen in the classroom. Information was procured through informal interviews beginning 1 week into the first intervention phase and continuing once every other week for the remainder of the study. Teachers and assistants were also asked the following questions in a questionnaire at the conclusion of the study: "Do you believe VSM helped the child?" "Did the implementation cause any disruption to the classroom routine?" "Did the child seem to enjoy watching the videos?" "Was the implementation distracting to other students?"

Treatment fidelity. Teachers were contacted every other day via email or direct contact and asked how the viewing went that day. Specifically, they were asked if there were any troubles with the technology, how the child responded to the video, and whether there were any changes in the routine of presentation. Each teacher reported that he or she had forgotten to show the video on one day to a child; thus, they were asked to add a day to these children's viewing so that all would have 5 days. The teachers' offices were very confined, and it would have been very crowded to have observers present; however, a recording of each child's first day of observation was made to gauge his or her reactions and attention to his or her videos. This also served to provide a check of whether teachers were following treatment protocol, and both teachers did so.

Results

Inter-Observer Agreement

Inter-observer agreement reached 90% for the two observers prior to the beginning of the baseline phase of the study. This criterion was achieved within 2 weeks of beginning observation of typically developing children. Likewise, the

same criterion was achieved with the third observer who began after the winter holidays. These high agreement rates were maintained throughout the study. Mean agreement for the first author and first-semester observer was 95%. The second observer had a 92% mean agreement.

VSM Intervention

All participants in this study appeared to make strong gains in the frequency of their social initiations upon introduction of the VSM movies. Maintenance of the behavior after withdrawal was also positive with no indications of regression. A summary of the mean rates of initiations across phases of the study is illustrated in Table 1. The results per data collection session are shown on the graph in Figure 1.

Taylor showed no change during his first three viewings of his movie. On the third day, he had four initiations, and the on fifth day, he had one, which was a direct verbal request to play.

It was apparent that the VSM intervention contributed to Ellen's improvement because she began going to a ball or Frisbee as soon as she entered the playground. This was the primary focus of her movie. She would then seek out others to play with her. Starting out with a zero baseline, Ellen averaged 2.8 initiations during intervention and maintained that rate.

Corey also showed an increase in initiations immediately after viewing his movie. Corey had the highest baseline rate, but it should be noted that all of his baseline initiations came about because someone took a toy from him resulting in him retrieving it from the other child. Almost all the baseline initiations across all four participants were related to retrieving a toy. On multiple occasions, Corey approached the observers and greeted them or asked questions. He did not do this during baseline.

Likewise, Jonathan showed strong gains beginning with his first viewing of the video. He began engaging in activities with others such as joining in on a group, going down slides, and playing games with a parachute. He did none of these during baseline. Like Ellen, the avoidance by peers, so often seen during baseline, was not nearly as apparent during the maintenance phase of the study.

Social Validity

The two teachers were supportive of the method and agreed that the dependent variable was an area of concern in the classroom and in the home. The skill was deemed important to the children's development. They reported almost no disruption to class routines and reported that the children enjoyed watching their videos. The reports about treatment efficacy were very positive with teachers and assistants reporting positive changes in class behaviors as well as the behaviors on the playground. This was not strictly related to initiations. Jonathan and Ellen were having problems with aggression in the classroom and it was reported that there

Table 1. Mean Initiations Within 15-Min Observations per Student per Stage of Study.

Child	Baseline	VSM	Maintenance
Taylor	.09	1.4	2.39
Ellen	.00	2.8	2.30
Corey	.39	2.60	2.79
Jonathan	.06	2.60	1.86

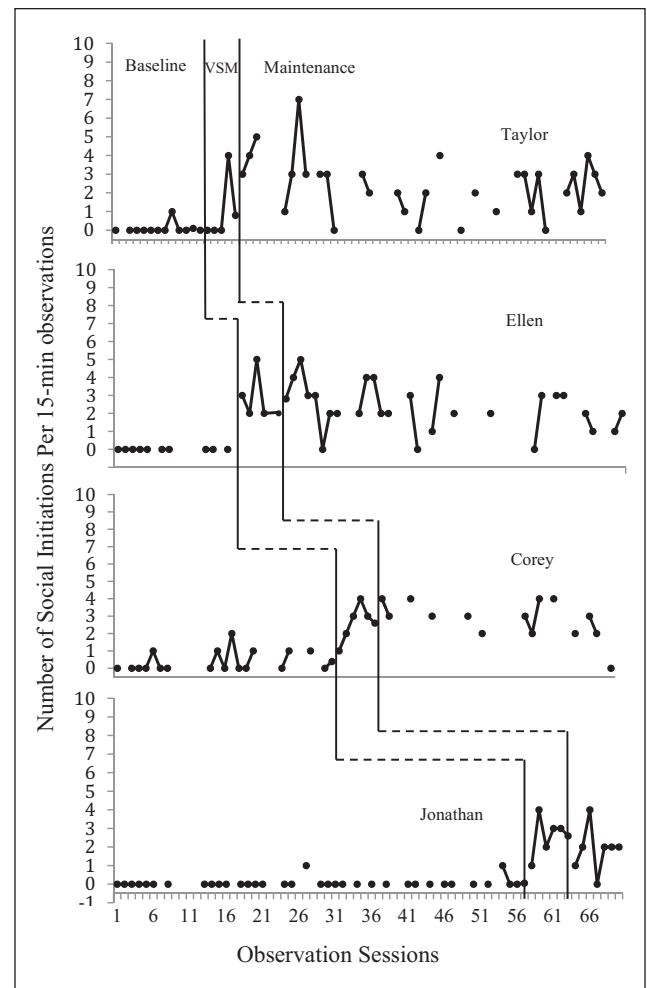


Figure 1. Daily observation data for each participant across the three phases of the study.

was a drop-off in aggression and other children were losing their fear of them.

In terms of practicality, the teachers stated that their time was so full that they did not know how they would make time for filming and editing. They suggested that therapists be trained to use it in their therapies. They also expressed pleasant surprise at how fast change was seen in all children. The fact that behavior could change so quickly with just 3-min viewings was seen by teachers as a justification for implementation.

Discussion

The implementation of a VSM intervention appeared to be successful across four study participants. Besides initiations, there were other changes in behavior of note both with the participants and peers. Ellen's interaction with the other children became more acceptable and the fleeing of peers on her approach ceased. She was seen in more intimate interactions with peers such as combing hair and holding hands.

Taylor's lack of progress following the first three VSM viewings was disappointing. However, one of the classroom assistants approached the first author on the playground and said, "This is incredible!" Asked to explain, she related that Taylor was now playing with friends and asking them to chase him. When told that the observers had seen nothing, she thought a second and suggested that we observe after another class left the playground (there were three classes on the playground at that time). As soon as that class left, Taylor left his "spot" and went up to two children and asked them to chase him. During baseline, observations were conducted randomly depending on the appearance of the children on the playground. However, Taylor was the first child observed during his intervention phase, mainly because of the curiosity of the researchers. During the next 2 days, observations were conducted after the other class departed. Thereafter, however, observations returned to a random format, and the presence of the other class no longer seemed to inhibit him. Based on teacher reports and the last 2 days of data collection during the intervention stage, we believe his mean score of 1.4 initiations is an underestimate of the effect of the interventions.

Taylor's results peaked shortly after watching his videos. We feel this was due to him seeking out peers to play. Over time, he discovered which of the children were receptive to his appeals for play resulting in fewer initiations being needed. Taylor was observed playing with others for long periods of time. The criteria set for what constituted an initiation, specifically the rules related to counting only the first initiation when there was repetition and the need for disengagement and re-engagement greatly affected his scores. He engaged in long activities with peers, and in that time, there were numerous verbal exchanges that would have counted as initiations had they not occurred in the same activities.

Corey's results were interesting because he became the focus of Taylor's requests for play. He often agreed and would chase Corey around the playground. Even though his participation in play increased, he still was not initiating. This changed upon viewing his movie. The day after beginning the viewing of his movie, he came up to the observers and initiated a conversation, asking what we were doing. He had not spoken to us during baseline. Most of Corey's baseline interventions involved reclaiming balls he was playing

with. Often the ball got away from him, and another child picked it up. His play during intervention changed considerably. He began initiating play with others. Up until then, he was a complete follower especially with Taylor. His time spent with Taylor decreased. He often said "no" when asked by Taylor to chase him. The number of different children he interacted with increased markedly.

It should be noted that Ellen and Jonathan had behavior plans. However, teachers were frustrated with lack of progress. All of the children had objectives for social behaviors in their IEPs, but again, little progress had been seen. Thus, the introduction of a 2.5-min video appeared to result in the mastery of behavior objectives very quickly: behaviors that had not been changed with other methods. The effects of the videos seemed to generalize across behaviors and settings. Positive changes were maintained by all participants; for two children for the majority of the school year. This may have been a result of a combination of classroom interventions and VSM, yet it was the moment when the videos were introduced that change was noted for all of the children.

There is enough evidence to justify the use of VSM with children with autism and other disabilities. Given the type of gains seen in the research, one could argue that VSM should be in the repertoire of all therapists and teachers. The results for social skills in children with autism younger than 4 years are not as positive. Much more research is needed. Research on other behaviors such as language and functional skills are lacking in the literature. It could be that children younger than 4 years could benefit in these areas.

Study Limitations

A threat to validity in single-subject design is small sample size, although this is compensated for by the opportunity to study individual differences and the staggered beginning of each intervention stage. Even though the researchers attempted to select children who were similar in abilities with similar demographics, individual differences within an n of 4 may constitute a threat, at least in terms of generalizability. Teachers and therapists at the school were aware of the study and it was obvious what types of behavior we were observing. All of the children had behavior plans. It is possible that staff at the institute reacted to the study by placing additional emphasis on the behavior plans or were influenced in their behavior toward the children in some manner.

Treatment fidelity was another threat in this study. Teachers were not observed during the child's viewing of the movie. Fidelity was monitored based mainly on teacher reports. Because teachers showed the videos in their small offices off the classrooms, observation was impractical. More careful control of treatment fidelity should be undertaken in future study.

Conclusion

The findings of the present study reflected findings of others working with young children with autism using VSM for social skill training (e.g., Bellini & Akullian, 2007; Buggey et al., 2012; Wert & Neisworth, 2003). The use of VSM to train social skills with 4-year-olds with autism was successful. This adds to the limited but consistent findings in training social skills with this population. VSM intervention was also successful for a young girl with DS within the 3- to 4-year age group. Children with DS have not been the focus of any VSM study to date. Hopefully, the success with Ellen will encourage others to investigate VSM as a tool for teaching these children.

One of the commonalities of this study and other studies (e.g., Bellini & Akullian, 2007; Buggey, 2012; Buggey et al., 2012) was statements by the teachers of surprise at the quickness and scope of behavior changes that they were seeing while saying that they did not have time to implement VSM. Given the dearth of effective teaching methods for children with ASD, or the length of time needed for some methods to be successful with these children, one would think that there would be a rush to embrace VSM. The belief that therapists should implement VSM probably has merit; however, it could also be the case that teachers are unfamiliar with how easy video editing has become. The benefits of class and child videos go beyond VSM into areas of scrapbooking, evaluation, creating portfolios, theater, and so forth. An effort should be made in preservice and inservice education training to make teachers aware of the benefits of video recording and editing, and direct instruction on the use of the equipment and software should be carried out. With these skills within a teacher's comfort zone, the likelihood of VSM implementation will probably improve significantly. This would make for a good study in the future.

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